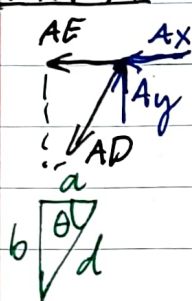


Pin A



$$AD_x = AD \cdot \frac{a}{d}$$

$$AD_y = AD \cdot \frac{b}{d}$$

$$\sin \theta = \frac{b}{d}$$

$$\cos \theta = \frac{a}{d}$$

$$\Sigma F_x = AE + AD_x = 0$$

$$AE + AD \frac{a}{d} = 0$$

$$\Rightarrow \Sigma F_x = AE + \left(\frac{d}{b} A_y \right) \frac{a}{d} = 0$$

$$\Sigma F_y = A_y - AD \cdot \frac{b}{d} = 0$$

$$\Rightarrow A_y \frac{d}{b} = AD \frac{b}{d} \cdot \frac{d}{b}$$

$$\boxed{\frac{d}{b} A_y = AD}$$

$$\Rightarrow AE + \frac{a}{b} A_y = 0$$

$$\Rightarrow AE = -\frac{a}{b} A_y$$

$$\boxed{AE = -\frac{a}{b} A_y}$$

Pin B



$$\sin \theta = \frac{b}{d}$$

$$\cos \theta = \frac{a}{d}$$

$$BC_y = BC \frac{b}{d}$$

$$BC_x = BC \frac{a}{d}$$

$$\Sigma F_y = -B_y - BC \frac{b}{d} = 0$$

$$\frac{d}{b} \cdot BC \frac{b}{d} = -B_y \cdot \frac{d}{b}$$

$$BC = -B_y \frac{d}{b}$$

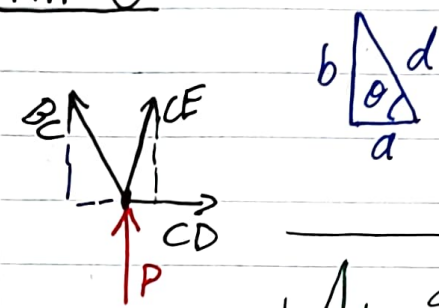
$$\Sigma F_x = B_x + BC \frac{a}{d} = 0$$

$$B_x + \left(-B_y \frac{d}{b}\right) \frac{a}{d} = 0$$

$$B_x - B_y \frac{a}{b} = 0$$

$$B_x = B_y \frac{a}{b}$$

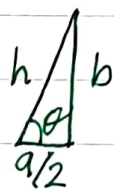
Pin C



$$\sin \theta = \frac{b}{d} \quad | \quad BC_x = BC \cdot \frac{a}{d}$$

$$\cos \theta = \frac{a}{d} \quad | \quad BC_y = BC \cdot \frac{b}{d}$$

$$h = \sqrt{(a/2)^2 + b^2}$$



$$\sin \theta = \frac{b}{h} \quad | \quad CE_x = CE \cdot \frac{a/2}{h}$$

$$\cos \theta = \frac{a/2}{h} \quad | \quad CE_y = CE \cdot \frac{b}{h}$$

$$\Sigma F_y = BC_y + CE_y + P = 0$$

$$\Rightarrow BC \frac{b}{d} + CE \frac{b}{h} + P = 0$$

$$BC = -B_y \frac{d}{b}$$

$$\frac{h}{b} \cdot CE \frac{b}{h} = -P - BC \frac{b}{d} \cdot \frac{h}{b}$$

$$CE = \frac{h}{b} (-P - BC \frac{b}{d})$$

$$CE = \frac{h}{b} (-P + (+B_y) (\frac{d}{b}) (\frac{b}{d}))$$

$$\boxed{CE = \frac{h}{b} (-P + B_y)}$$

$$\Sigma F_x = CD + CE_x - BC_x = 0$$

$$\boxed{CD = -B_y \frac{a}{b} - \frac{a/2}{b} (B_y - P)}$$

$$CD = BC_x - CE_x$$

$$CD = BC \cdot \frac{a}{d} - CE \frac{a/2}{h}$$

$$CD = (-B_y \frac{d}{b}) \frac{a}{d} - (\frac{h}{b} (-P + B_y)) \frac{a/2}{h}$$

Pin D



$$\begin{aligned} \sin \theta &= \frac{b}{d} \quad \checkmark & AD_x &= AD \frac{a}{d} \\ \cos \theta &= \frac{a}{d} \quad \checkmark & AD_y &= AD \frac{b}{d} \end{aligned}$$



$$\begin{aligned} \sin \theta &= \frac{b}{h} \quad \checkmark & DE_x &= DE \frac{a/2}{h} \\ \cos \theta &= \frac{a/2}{h} \quad \checkmark & DE_y &= DE \frac{b}{h} \end{aligned}$$

$$\Sigma F_y = DE_y + AD_y - P = 0$$

$$DE_y = P - AD_y$$

$$AD = \frac{d}{b} A_y$$

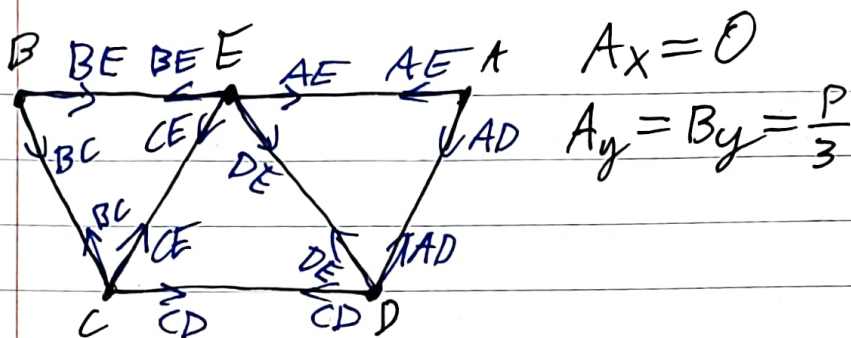
$$DE \frac{b}{h} = P - AD \frac{b}{d}$$

$$DE = \frac{h}{b} \left(P - AD \frac{b}{d} \right)$$

$$DE = \frac{h}{b} \left(P - \left(A_y \frac{d}{b} \right) \frac{b}{d} \right)$$

$$\boxed{DE = \frac{h}{b} (P - A_y)}$$

$$\Sigma F_x = AD \frac{a}{d} - CD - DE \frac{a/2}{h} = 0$$



$$a = 0.4 \text{ m}$$

$$b = 0.3 \text{ m}$$

$$d = \sqrt{a^2 + b^2}$$

$$d = 0.5 \text{ m}$$

$$h = \sqrt{a^2 + b^2}$$

$$h = 0.3605551275$$

$$^1 AD = \frac{d}{b} A_y = \frac{d}{b} \cdot \frac{P}{3}$$

$$^2 AE = -\frac{a}{b} A_y = -\frac{a}{b} \cdot \frac{P}{3}$$

$$^3 DE = \frac{h}{b} (P - A_y) = \frac{h}{b} (P - \frac{P}{3})$$

$$^4 CD = -B_y \frac{a}{b} - \frac{a/2}{b} (B_y - P) = -\frac{P}{3} \cdot \frac{a}{b} - \frac{a/2}{b} (\frac{P}{3} - P)$$

$$^5 CE = \frac{h}{b} (-P + B_y) = \frac{h}{b} (-P + \frac{P}{3})$$

$$^6 BC = -B_y \frac{d}{b} = -\frac{P}{3} \cdot \frac{d}{b}$$

$$^7 BE = B_y \frac{a}{b} = \frac{P}{3} \cdot \frac{a}{b}$$